

Keplerian Radial-Velocity Fitting for Single-Planet Exoplanet Parameter Estimation

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Abstract

The radial-velocity (RV) method remains one of the foundational techniques for detecting and characterizing extrasolar planets. We present a compact, reproducible pipeline for estimating the orbital parameters of a single companion from Doppler time series. The forward model is the standard Keplerian RV curve parameterized by semi-amplitude K , period P , eccentricity e , argument of periastron ω , time of periastron t_0 , and systemic velocity γ . Parameters are recovered with a global differential-evolution search followed by local nonlinear least-squares refinement. From the fitted amplitude and an assumed stellar mass we derive a minimum planet mass and orbital semi-major axis under the edge-on inclination limit. Synthetic experiments recover planted values of K and P to within a few percent at residual RMS of $\sim 2 \text{ m s}^{-1}$. Application to an observational Doppler-shift series yields a short-period, nearly circular solution with $P \approx 4.27 \text{ d}$, $K \approx 56.7 \text{ m s}^{-1}$, and $M_p \sin i \approx 0.49 M_J$. A velocity-Verlet two-body integration is used only for post-fit orbit visualization.

1 Introduction

The discovery of 51 Pegasi b established Doppler spectroscopy as a practical route to detecting planets around solar-type stars [1, 2]. In the RV method, the star and planet orbit their common barycenter; the line-of-sight component of the stellar motion appears as a periodic Doppler shift in the stellar spectrum. Modern surveys and modeling frameworks treat this signal with analytic Keplerian orbits and statistically motivated fitting procedures [3–5].

For a single planet on a Keplerian orbit, the stellar radial velocity is fully specified by a small set of orbital elements [6, 7]. The semi-amplitude K encodes the planet-to-star mass ratio and orbital scale, while P and e control the periodicity and waveform shape. Because the likelihood surface can be multimodal—especially when eccentricity and phase are free—global optimization or Bayesian sampling is often preferred over purely local gradient methods [8, 9].

This work implements a transparent single-planet RV estimator intended for reproducible analysis and pedagogical demonstration. The pipeline (i) converts Doppler shifts to radial velocities, (ii) fits a Keplerian model with differential evolution plus least-squares polish, and (iii) derives physical parameters and a barycentric orbit visualization. We validate the method on synthetic data with known ground truth and apply it to an observational Doppler time series.

2 Methods

2.1 Doppler shift and radial velocity

Spectroscopic observations may be reported as a fractional Doppler shift $z = \Delta\lambda/\lambda$. For a source with line-of-sight speed v relative to the observer, the relativistic Doppler formula is

$$z = \sqrt{\frac{1+\beta}{1-\beta}} - 1, \quad \beta = \frac{v}{c}, \quad (1)$$

where c is the speed of light [10]. Inverting this relation yields

$$v = c \frac{(1+z)^2 - 1}{(1+z)^2 + 1}. \quad (2)$$

Throughout, positive v denotes recession.

2.2 Keplerian radial-velocity model

We adopt the standard single-planet RV model [3, 7]:

$$v_r(t) = \gamma + K \left[\cos(\omega + \nu(t)) + e \cos \omega \right], \quad (3)$$

where γ is the systemic velocity, K is the RV semi-amplitude, e is the eccentricity, ω is the argument of periastron, and $\nu(t)$ is the true anomaly.

The true anomaly is obtained from the mean anomaly

$$M = \frac{2\pi}{P} (t - t_0) \quad (4)$$

by solving Kepler's equation

$$E - e \sin E = M \quad (5)$$

for the eccentric anomaly E , then converting

$$\tan \frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2} \quad (6)$$

[6].

For an edge-on orbit ($\sin i = 1$), the semi-amplitude is related to the component masses and period by

$$K = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{M_p}{(M_\star + M_p)^{2/3}} \frac{1}{\sqrt{1-e^2}}, \quad (7)$$

where G is Newton's constant. Given K , P , e , and $M_\star$, we solve Eq. (7) for M_p and obtain the semi-major axis from Kepler's third law,

$$a^3 = \frac{G(M_\star + M_p)}{4\pi^2} P^2. \quad (8)$$

When $\sin i < 1$, the mass inferred from Eq. (7) is a lower limit $M_p \sin i$.

2.3 Parameter estimation

Let $\{t_j, v_j, \sigma_j\}$ denote the observed times, radial velocities, and uncertainties. We minimize the weighted residual sum of squares

$$\chi^2(\boldsymbol{\theta}) = \sum_j \left(\frac{v_j - v_r(t_j; \boldsymbol{\theta})}{\sigma_j} \right)^2, \quad (9)$$

with parameter vector $\boldsymbol{\theta} = (K, P, e, \omega, t_0, \gamma)$.

Because χ^2 is nonconvex in the orbital elements, we first perform a global search with differential evolution [9, 11], which maintains a population of candidate solutions and updates them by mutation and crossover. The best candidate is then polished with a bounded nonlinear least-squares solver [11]. Search bounds are constructed from the observation baseline and the observed velocity span, with $0 \leq e \leq 0.85$ to avoid near-parabolic degeneracies.

2.4 Orbit visualization

After fitting, we construct barycentric initial conditions consistent with the derived a , e , and component masses, and integrate the Newtonian two-body problem with the velocity Verlet algorithm [12]. This integration is used solely for visualization of the reflex orbit; it is not part of the likelihood evaluation.

3 Data

3.1 Synthetic validation set

To quantify recovery accuracy we generate a synthetic RV series for a Jupiter-class companion around a $\sim 1.08 M_\odot$ star with planted parameters

$$P = 147.5 \text{ d}, \quad K \approx 30.5 \text{ m s}^{-1}, \quad e = 0.19, \quad M_p = 0.82 M_J.$$

Observation times are drawn irregularly over a 420 d window. Independent Gaussian noise of standard deviation 2.2 m s^{-1} and an additional jitter term of 0.7 m s^{-1} are added to each sample.

3.2 Observational Doppler series

We also analyze an observational Doppler-shift time series for a $1.11 M_\odot$ host (dataset `star0`). Times are recorded in seconds and converted to days; Doppler shifts are converted to radial velocities as described above. In the absence of published per-point uncertainties we adopt a uniform $\sigma = 1 \text{ m s}^{-1}$ for weighting. Three additional hosts with stellar masses 1.39, 2.30, and $1.34 M_\odot$ are available in the same catalog for follow-up analysis.

4 Results

4.1 Synthetic recovery

Table 1 compares planted and recovered parameters. The period and semi-amplitude are recovered to within $\sim 3\%$, and the residual RMS is 2.11 m s^{-1} , consistent with the injected noise level. Figure 1 shows the observations, the best-fit Keplerian curve, the ground-truth curve, and the corresponding barycentric orbit.

Table 1: Synthetic validation: planted truth versus recovered fit.

Quantity	Truth	Fit	Relative error
P (d)	147.5	147.16	0.2%
K (m s^{-1})	30.51	31.35	2.8%
e	0.190	0.221	—
M_p (M_J)	0.820	0.836	2.0%
RMS (m s^{-1})	—	2.11	—

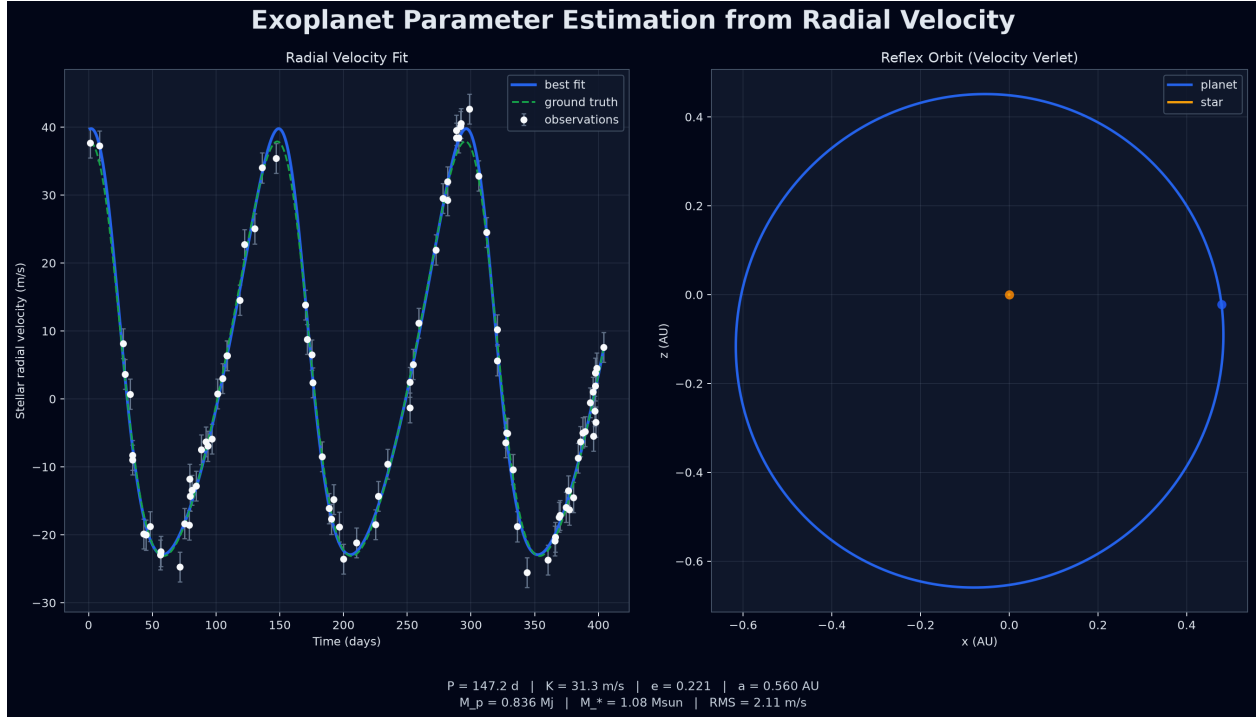


Figure 1: Synthetic radial-velocity fit. Left: noisy observations (points), best-fit Keplerian model (solid), and planted ground truth (dashed). Right: barycentric star–planet orbit integrated with velocity Verlet from the fitted elements.

4.2 Observational fit for star0

Fitting the `star0` Doppler series yields a short-period, nearly circular solution (Table 2; Fig. 2). The low eccentricity and coherent phase coverage over ~ 14 d produce a stable global minimum. Under the edge-on assumption the companion mass is $M_p \sin i \approx 0.49 M_J$ at $a \approx 0.053$ AU.

Table 2: Best-fit Keplerian parameters for the **star0** Doppler series ($M_\star = 1.11 M_\odot$).

Parameter	Value
P (d)	4.269
K (m s^{-1})	56.694
e	0.002
γ (m s^{-1})	9.398
a (AU)	0.053
$M_p \sin i$ (M_J)	0.485
RMS (m s^{-1})	5.23

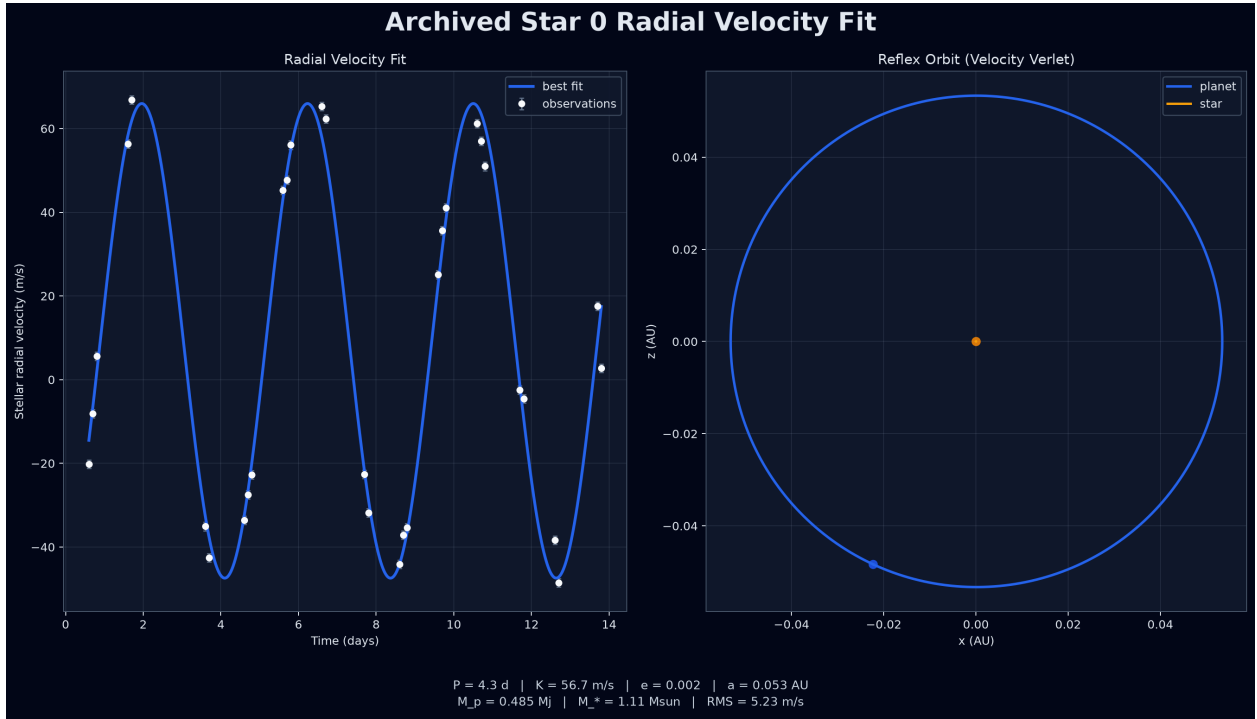


Figure 2: Keplerian fit to the **star0** Doppler observations. Left: radial-velocity time series and best-fit model. Right: corresponding barycentric orbit.

5 Discussion

The synthetic experiment confirms that differential evolution followed by local refinement is sufficient to recover the dominant Keplerian parameters when the noise is Gaussian and the sampling covers multiple orbital cycles [3, 9]. Eccentricity is the most weakly constrained element in our synthetic run, as expected when the signal-to-noise ratio is modest and the true e is moderate [8].

For **star0**, the near-zero eccentricity and short period are characteristic of hot-Jupiter systems first revealed by RV surveys [1]. Several caveats apply. First, the adopted uncertainties are uniform and may not reflect the true error budget, so the formal χ^2 should not be over-interpreted. Second, the mass we report is $M_p \sin i$; without an independent inclination constraint the true mass may be

larger. Third, the model assumes a single planet and a constant systemic velocity, omitting stellar activity, instrumental offsets, and additional companions [5].

Natural extensions include periodogram-based period initialization [4], multi-planet Keplerian models [3], and Bayesian posterior sampling for full uncertainty quantification [8].

6 Conclusions

We have described a Keplerian radial-velocity pipeline for single-planet parameter estimation. The method combines an analytic RV forward model, differential-evolution global search, and least-squares refinement, with derived planet mass and semi-major axis under the edge-on limit. Synthetic tests recover K and P to within a few percent. Application to an observational Doppler series yields a coherent short-period, low-eccentricity solution. The accompanying open-source implementation provides a reproducible basis for further methodological development.

Software

The analysis is implemented in Python using NumPy, SciPy [11], and Matplotlib. Source code and example scripts are available in the project repository accompanying this manuscript.

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